

$\mu \in \mathbb{R}^3$
center (position)

μ (3D center)

$q = (r, x, y, z)$ normalize, then build $R(q)$
quaternion

rotation, 3×3

$$\Sigma = R S^2 R^\top$$

3×3 symmetric, PSD

S^2 and R combine into Σ

$\mathbf{s} \in \mathbb{R}^3$
log-scale

exp

$$S = \text{diag}(e^{s_x}, e^{s_y}, e^{s_z})$$